



Cell-Subtype Changes in Adenosine Receptor Expression in Schizophrenia

Smita Sahay, Emily A. Devine, Robert E. McCullumsmith, Sinead M. O'Donovan

Department of Neurosciences, The University of Toledo College of Medicine and Life Sciences, Toledo, OH, USA

INTRODUCTION

Schizophrenia is a debilitating neuropsychiatric disorder affecting approximately 0.6% of the U.S. population and is characterized by altered thought processes, hallucinations, and delusions. Dysregulation of the glutamate and dopamine neurotransmitter systems are implicated in the pathophysiology of schizophrenia.

The adenosine system is an important neuroregulatory system in the brain that modulates glutamate and dopamine signaling via the ubiquitously expressed adenosine receptors. However, adenosine A₁ and A_{2A} receptor (A₁R and A_{2A}R) expression is poorly understood in specific cell-subtypes in Frontal cortical brain regions implicated in this disorder.

In this study, we assayed A₁R and A_{2A}R mRNA expression by qPCR in enriched populations of pyramidal neurons, isolated from postmortem anterior cingulate cortex (ACC) tissue from age and sex-matched schizophrenia (n=20) and control (n=20) subjects, using laser microdissection (LMD).

METHODS

Enriched populations of pyramidal neurons were isolated from the postmortem ACC tissue of schizophrenia (n = 20) and control (n = 20) subjects using (LMD).

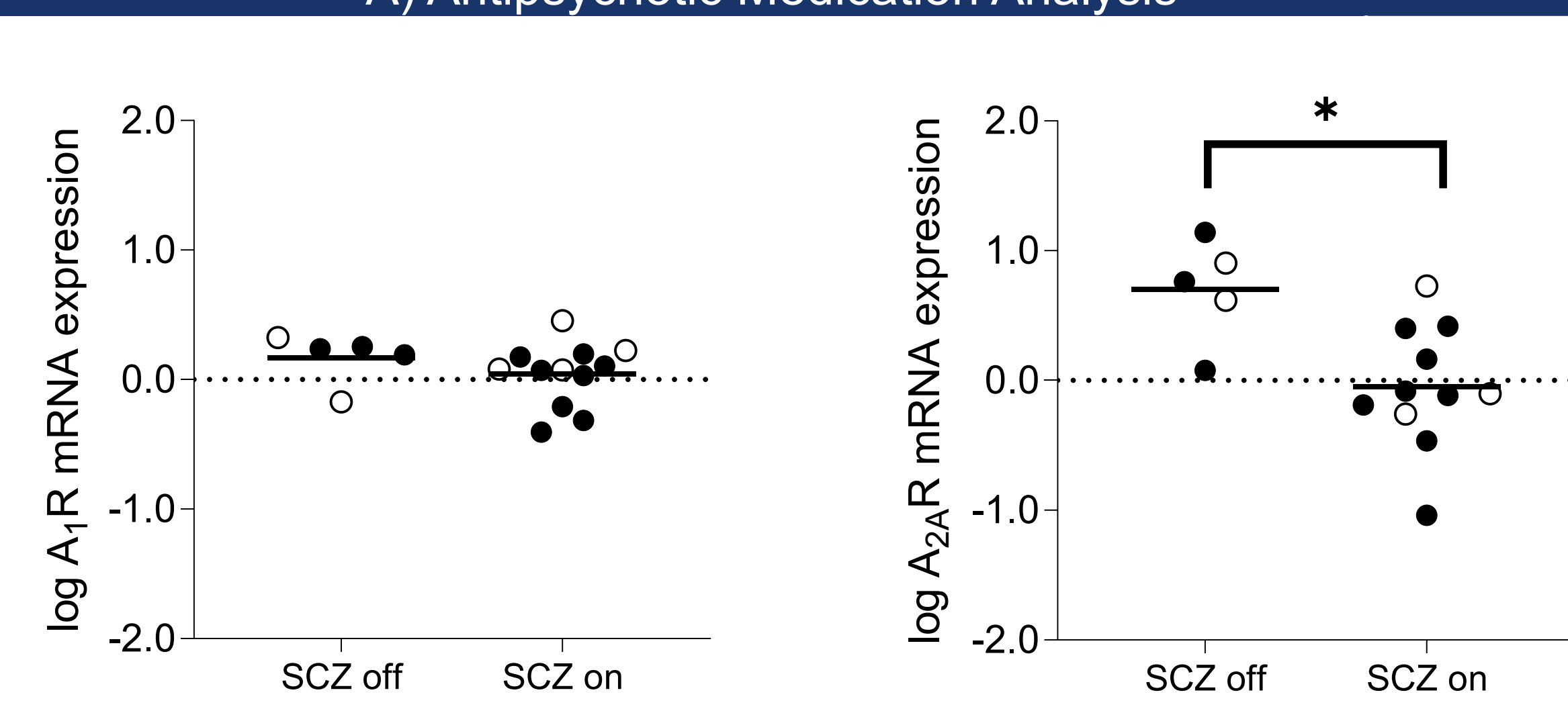
Expression of A₁Rs and A_{2A}Rs was assayed using quantitative polymerase chain reaction (qPCR). A₁R and A_{2A}R expression was quantified and analyzed based on sex, clinical dementia rating (CDR) scores, and antipsychotic use.

ACKNOWLEDGMENTS

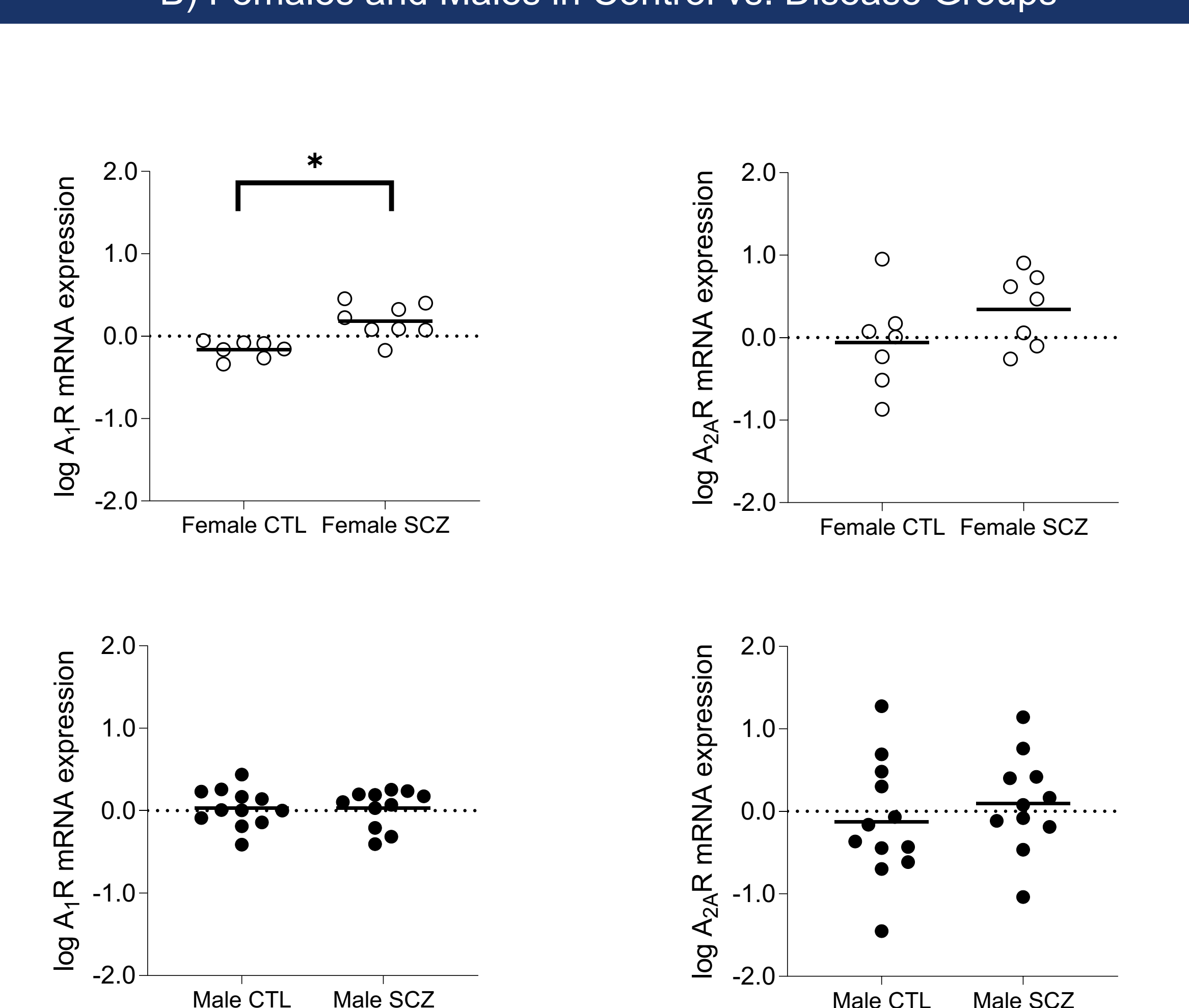
This research was funded by the National Institute of Mental Health (MH107487) and the American Foundation of Suicide Prevention (YIG-1-139-20).

RESULTS

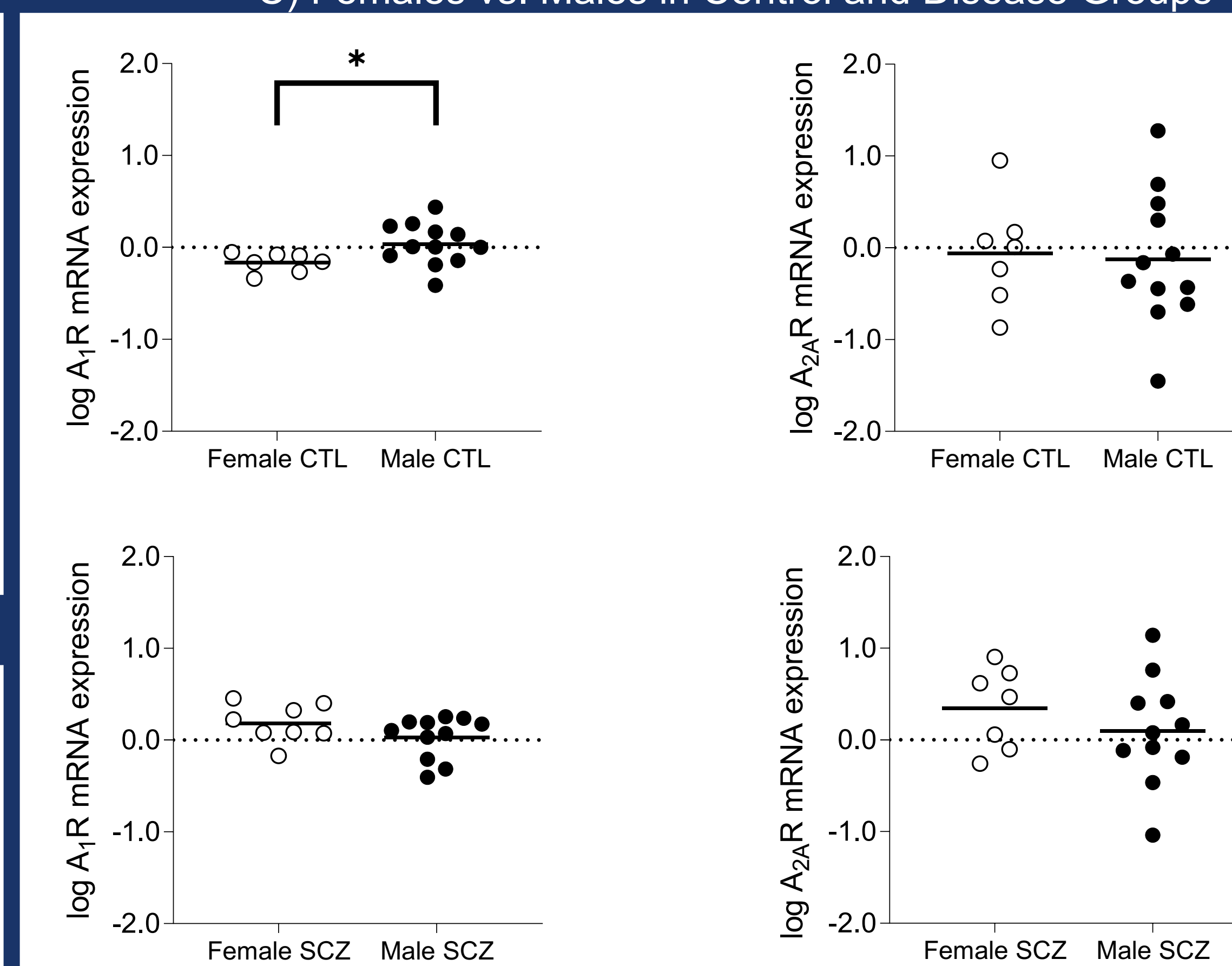
A) Antipsychotic Medication Analysis



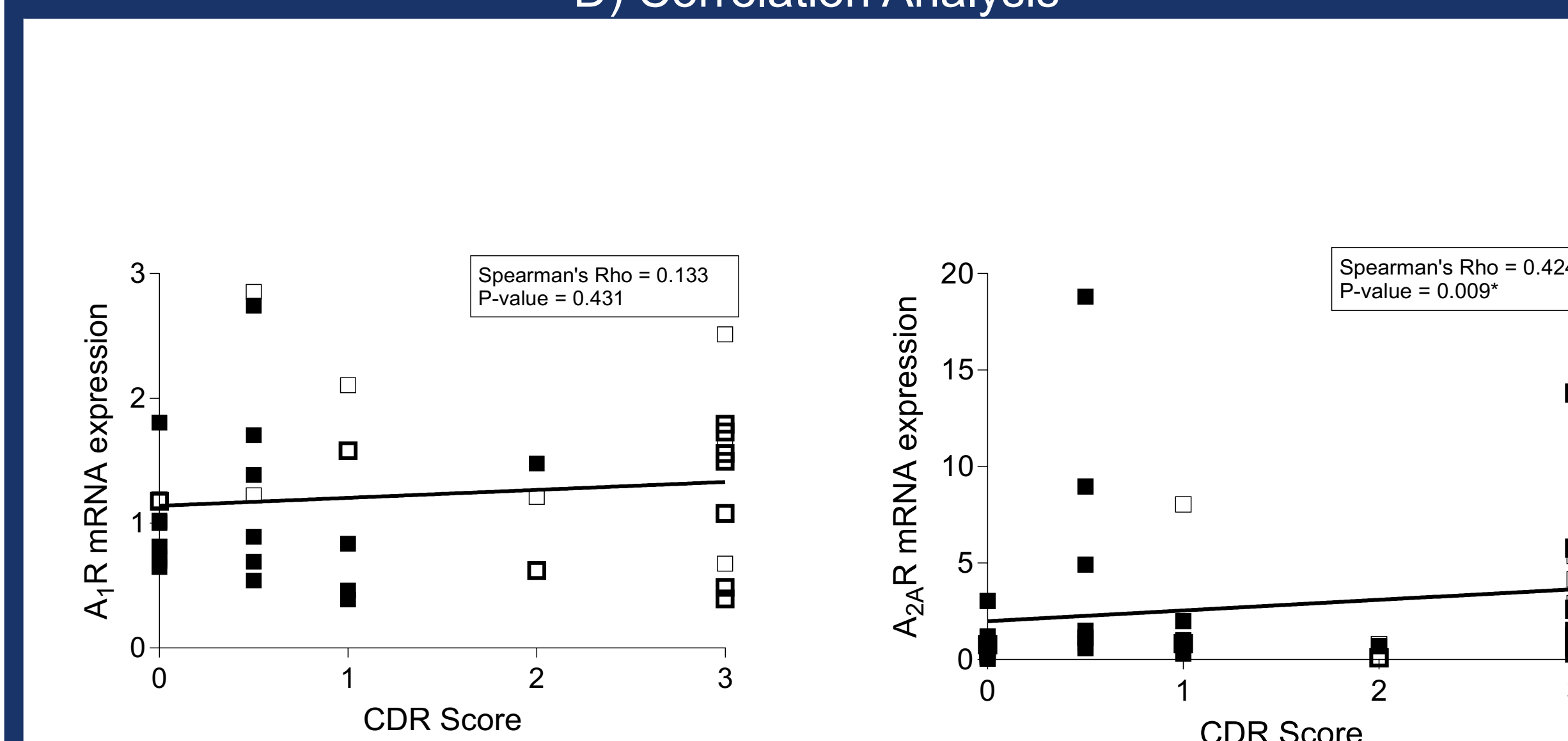
B) Females and Males in Control vs. Disease Groups



C) Females vs. Males in Control and Disease Groups



D) Correlation Analysis



A) A_{2A}R mRNA expression was significantly increased in schizophrenia subjects who were off antipsychotic medication (ANCOVA: $F_{(1,12)}=6.444$, $p=0.026$).

B) A₁R expression was significantly increased in female schizophrenia subjects compared to female controls ($t_{(13)}=-4.008$, $p=0.001$).

C) A₁R expression was significantly decreased in female controls compared to male controls ($t_{(17)}=2.137$, $p=0.047$).

D) There was a significant positive association between dementia severity and A_{2A}R mRNA expression (Spearman's $r=0.424$, $p=0.009$).

In Figures A-C, open circles indicate females, closed circles indicate males. In Figure D, open squares indicate schizophrenia, closed circles indicate controls.

DISCUSSION

- The results from our medication analysis indicate that A_{2A}R expression may be normalized by antipsychotic treatment; however, a limitation of this analysis is the low number of subjects (n = 5) off antipsychotic medications.
- A₁R expression was increased in female schizophrenia subjects compared to female controls, indicating sex-specific increases in A₁R expression in this study.

- A₁R expression was significantly decreased in female controls compared to male controls, suggesting sex differences in basal A₁R expression.
- There is likely a relationship between dementia severity and A_{2A}R mRNA expression, as subjects with more severe dementia have increased expression, regardless of sex.
- Overall, these results provide novel insights into the cell-subtype specific expression of adenosine receptors in the ACC in schizophrenia and suggest that changes in receptor mRNA expression may be sex-dependent and associated with dementia in these subjects.